ARNAV ANAND

Computer Vision & Machine Learning Engineer

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SUMMARY

Computer Vision & ML engineer with 4+ years shipping production vision systems end to end — annotation standards and labeling tooling through model training, threshold calibration, edge benchmarking, and monitored deployment. Own the data-quality and calibration layer of a large-scale imagery platform at GenLogs; previously shipped real-time video analytics at Wobot.ai. Two peer-reviewed publications.

EXPERIENCE

GenLogs **Mar 2026 – Present**

Computer Vision Engineer *Dallas–Fort Worth, TX*

- Built and deployed the team's CV data-sampling and curation dashboard (local + production, VPN-gated) with five pipeline flows — sampling, class backfill, hard-case ingest, model-assisted drafting, and upload — a single path from raw frames to labeled datasets.
- Own the canonical annotation guidelines and object taxonomy; designed and led a taxonomy migration across a multi-thousand-image dataset using idempotent, content-addressed scripts written back into the annotation platform.
- Integrated custom detection models into the annotation platform for pre-labeling where no native path existed (model export, offline inference, review-mode configuration); benchmarked the draft labeller on representative edge hardware (ms/frame, p50/p95) to guide resolution and deployment choices.
- Lead similarity-threshold calibration for embedding-based image search: precision-first, uniformly normalized method with a reproducible analysis + refresh pipeline and a SQL-warehouse inference prototype handed off to data engineering.
- Closed the loop between production failures and evaluation by ingesting field-reported hard cases into frozen, provenance-tagged test sets; cut false positives in a production classifier via per-camera confidence-threshold analysis.

Wobot.ai **Mar 2021 – May 2023**

Computer Vision Engineer (Computer Vision Intern, Mar–Jun 2021) *Foster City, CA · Remote*

- Designed, trained, and shipped end-to-end real-time video analytics pipelines (ingest → preprocess → train → deploy) in TensorFlow/PyTorch with automated CI/CD on AWS for production customer workloads.
- Built object detection, multi-object tracking, pose-estimation, and ROI-breach solutions plus FastAPI services for real-time inference and model versioning; scaled ROI logic across multiple regions and clouds (AWS, Azure).
- Ran production on Docker/Kubernetes with Prometheus, Grafana, and Loki for system and model health; optimized throughput and latency with ONNX/TensorRT; extended the in-house CV library, led code review, and mentored interns.

Gritt Robotics **Sep 2024 – Dec 2024**

Software Engineer Intern, Computer Vision *San Francisco Bay Area · Remote*

- Lifted per-class mAP by 15% for a real-time perception stack by fine-tuning Detectron2 (Mask R-CNN / PointRend) on curated synthetic + real data and grid-searching score/NMS thresholds into hardened production configs.
- Built camera-calibration health checks with ground-truth vs prediction visualizations; integrated calibrated feeds into the on-device inference pipeline with performance monitoring and wrote calibration/evaluation SOPs.

International Institute of Information Technology, Hyderabad **Jan 2022 – Jul 2022**

Research Engineer, IHub-Data *Remote*

- Raised a PyTorch pothole-detection model from 83% to 95% accuracy through systematic hyperparameter tuning, experiment tracking, and augmentation; built Spark Streaming → PostgreSQL pipelines with live Grafana heatmaps and alerting for statewide inference data.

Delhi Technological University — Biometrics Lab **May 2020 – Dec 2020**

Research Intern *New Delhi, India*

- Co-authored two peer-reviewed face anti-spoofing papers — "A deep face spoof detection framework using multi-level ELBPs and stacked LSTMs" (Springer SIViP) and "Face anti-spoofing by spatial fusion of colour texture features and deep features" (IEEE ICISS 2020).

SKILLS

CV / ML: PyTorch, TensorFlow, Detectron2, OpenCV, object detection & tracking, instance segmentation, pose estimation, image embeddings & similarity search, threshold calibration, dataset curation & annotation tooling, model evaluation

MLOps / Infra: ONNX Runtime, TensorRT, NVIDIA Triton, Docker, Kubernetes, AWS, Azure, CI/CD, MLflow, Prometheus/Grafana/Loki, edge benchmarking

Data / Backend: Python, C++, SQL (Snowflake, PostgreSQL), Spark Streaming, FastAPI, Linux, Git

EDUCATION

The University of Texas at Dallas — M.S. Computer Engineering **Aug 2023 – May 2025**

Guru Gobind Singh Indraprastha University (USICT) — B.Tech. Information Technology **2017 – 2021**